

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

ACTU 301: LIFE CONTINGENCIES 1 (3 credits)

Question 1

- i) Given the survival function $S(t) = e^{-(\beta t)^\gamma}$, if $t \geq 0$ where $\beta > 0$ and $\gamma > 0$, obtain the probability density function of the lifetime random variable and the hazard function.
- ii) Assuming the lifetime of a person is given by an exponential distribution with an expected value of 80 years, find the probability the person will survive to age 50. What is the value of the hazard function at this age?
- iii) Prove that if the hazard function of a continuous distribution is constant, the distribution is exponential.
- iv) Establish the relationship between the force of mortality of a person age x and the given survival function $S(t)$.

Question 2

- i) Let

$$F(x) = 1 - \left(1 - \frac{x}{120}\right)^{\frac{1}{6}}, 0 \leq x \leq 120.$$

Determine the probability that a life aged 35 dies before the age of 55.

- ii) Consider an age-at-death random variable X with survival distribution defined by

$$s(x) = \frac{1}{10} (100 - x)^{\frac{1}{2}}, 0 \leq x \leq 100.$$

(a) Explain why this is a suitable survival function.

(b) Find the corresponding expression for the cumulative probability function.

(c) Compute the probability that a newborn with survival function defined above will die between the ages of 65 and 75.

iii) Given $\mu_{x+t} = 0.002t, \quad t \geq 0.$

(a) Is μ_{x+t} a valid function for the force of mortality of (x)?

(b) Find $S_x(t)$.

(c) Find $f_x(t)$.